

$$\begin{array}{ccc}
\mathcal{D} \left(F \left(X \otimes_{\mathcal{C}} \iota F Y \right), A \right) & \xrightarrow{\mathcal{D}(F(X \otimes_{\mathcal{C}} \eta_Y), A)} & \mathcal{D} \left(F \left(X \otimes_{\mathcal{C}} Y \right), A \right) \\
\cong \downarrow & & \downarrow \cong \\
\mathcal{C} \left(X \otimes_{\mathcal{C}} \iota F Y, \iota A \right) & \xrightarrow{\mathcal{C}(X \otimes_{\mathcal{C}} \eta_Y, \iota A)} & \mathcal{C} \left(X \otimes_{\mathcal{C}} Y, \iota A \right) \\
\cong \downarrow & & \downarrow \cong \\
\mathcal{C} \left(X, [\iota F Y, \iota A]_{\mathcal{C}} \right) & \xrightarrow{\mathcal{C}(X, [\eta_Y, \iota A]_{\mathcal{C}})} & \mathcal{C} \left(X, [Y, \iota A]_{\mathcal{C}} \right).
\end{array}$$